

Optimal blue excitation wavelength for nitrogen-vacancy optically detected magnetic resonance at 120 GPa

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Abstract

Nitrogen-vacancy (NV) centers in the culet of a diamond anvil cell are the reference local probe of magnetism in the megabar regime, where their usefulness is limited not by whether optically detected magnetic resonance (ODMR) survives compression but by how fast a given sensitivity is reached. We show that at 120 GPa the sensitivity is controlled almost entirely by the excitation wavelength, a choice usually inherited from ambient-pressure practice. Using a rate-equation model of the lock-in sensitivity $\eta \propto \Delta\nu/(C\sqrt{R})$ anchored to the measured and calculated photophysics of the NV center up to 120 GPa, we find that the zero-phonon-line blue shift and the strengthened electron-phonon coupling place the optimum at $\lambda_{\text{opt}} = 475$ nm, while the ground-state photoionization threshold imposes a hard blue boundary at 405 nm. Both conventional choices fall outside the window: 532 nm costs a factor 5.2 and 405 nm a factor 4.2. Within the window the steady-state charge state is wavelength independent to 0.2%, because excited-state ionization and recombination both scale with the absorption cross section and cancel, so the fixed-power optimum coincides with the Franck-Condon maximum—a prediction the proposed two-beam calibration can falsify. The 5% tolerance window, 463–486 nm, contains the commercial 473 nm line.

Keywords: nitrogen-vacancy center, diamond anvil cell, optically detected magnetic resonance, high-pressure magnetometry, photoionization, excitation wavelength

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1. Introduction

The search for superconductivity in hydrogen-rich compounds, initiated by Ashcroft’s proposal that metallic hydrogen and hydrogen-dominant alloys should be high-temperature superconductors [1, 2], has produced the highest critical temperatures on record: 203 K in H_3S at 155 GPa [3] and 250–260 K in LaH_{10} near 170 GPa [4, 5]. A second family of clathrate superhydrides operates at markedly lower pressure: CeH_9 is synthesized at 80–100 GPa and superconducts with T_c up to 115 K below one megabar [6, 7], and the substitutional alloy $(\text{La,Ce})\text{H}_9$ reaches 148–178 K over 97–172 GPa [8]. These materials are the most attractive near-term targets precisely because their working pressures, clustered around 120 GPa, are compatible with a well-characterized diamond anvil cell (DAC) rather than with an extreme, low-yield one.

What has not kept pace is the *evidence*. A sample in a DAC at 120 GPa occupies a culet of a few tens of micrometers and a volume of order picoliters, so conventional bulk magnetometry cannot isolate its diamagnetic response from the anvil and gasket background [9]. Resistive signatures alone have repeatedly proved insufficient to settle claims of superconductivity in this field, which has made a direct, local, and spatially resolved measurement of magnetic flux expulsion the decisive experiment rather than a confirmatory one.

NV centers implanted a few tens of nanometers beneath the culet surface supply exactly this measurement. Because the sensor layer is fabricated *inside* the anvil, it sits micrometers from the sample and survives the same pressure the sample does; ODMR then reports the local magnetic field and the local stress tensor simultaneously [10, 11, 12, 13]. This capability has been pushed from the 30 GPa scale to 130 GPa [14], to megabar pressures [15, 16], and most recently to 240 GPa [17]. Its highest-profile application to date is the imaging of the Meissner effect and of flux trapping in CeH_9 at megabar pressures, which resolved the superconducting regions and showed that they are strongly inhomogeneous on the micron scale [18]; comparable measurements have since been reported for LaH_{10} [19], for conventional superconductors under pressure [20], and for pressurized nickelates [21].

That last result identifies the bottleneck. The scientific content of these experiments is carried by *maps*—the spatial texture of the screening field, the geometry of trapped flux—and by their evolution through T_c and with pressure. A map is a raster of ODMR spectra, so the total measurement time

scales as the number of pixels divided by the square of the magnetic sensitivity η_B [22]. In a regime where a single successful loading takes weeks to prepare and can be lost in one anvil failure, sensitivity is not a figure of merit but the currency in which spatial resolution, field-of-view, and the number of accessible (P, T) points are all purchased. The relevant question for NV sensing at 120 GPa is therefore not whether ODMR survives compression—recent photophysical work shows convincingly that it does [23, 24]—but how to make it as sensitive as possible there.

The answer is constrained by the fact that the NV center at 120 GPa is optically a different emitter from the one at ambient pressure, in three ways that a recent combined *ab initio* and experimental study has now quantified [23]. First, the NV^- zero-phonon line (ZPL) blue-shifts by more than 0.4 eV, from 1.945 eV to $\simeq 2.35$ eV (529 nm). The 532 nm line, which at ambient pressure excites comfortably into the phonon sideband, therefore lands *at* the absorption onset at 120 GPa, where the cross section has collapsed. Excitation must move into the blue; this is a constraint, not a preference. Second, the electron-phonon coupling strengthens substantially—the absorption Huang-Rhys factor of the Jahn-Teller-inactive modes rises from 3.08 to 4.61 and the absorption Debye-Waller factor falls from 2.2% to 0.36%—so the absorption envelope both broadens and moves its weight to higher energy, placing its maximum near 475 nm. Third, the photoionization thresholds shift upward as well: $\text{IP}(^3\text{A}_2)$ rises from 2.68 eV to 3.06 eV and $\text{IP}(^3\text{E})$ from 1.16 eV to 1.63 eV [23, 25]. Since 3.06 eV corresponds to 405 nm, the 405 nm diode line that is standard for charge-state manipulation sits exactly on the ground-state ionization edge at 120 GPa: it is the last wavelength before one-photon ionization out of $^3\text{A}_2$ switches on, and everything bluer converts NV^- directly into NV^0 .

The three shifts do not act in the same direction, and that is what makes the problem nontrivial. Moving the excitation bluer follows the displaced absorption band and buys photon rate; moving it bluer also approaches the ground-state ionization edge, depletes the steady-state NV^- fraction f_- , and dilutes the ODMR contrast with unpolarized NV^0 fluorescence. The lock-in sensitivity, $\eta \propto \Delta\nu/(C\sqrt{R})$, weighs these against each other with different powers—linearly in contrast, as a square root in photon rate—so where the balance falls is a quantitative question and not one that can be settled by inspecting the absorption spectrum. Ref. [23] recognizes the issue and states the qualitative guideline that the excitation should be tuned to follow the ZPL shift, but stops short of the quantitative question that an experimenter

has to answer when specifying a laser: *which* wavelength, to what tolerance, and whether that choice depends on the optical power—a question with direct consequences in a DAC, where the culet is small, the sample is metallic and absorbing, and the usable power is capped by heating rather than by the laser.

Here we answer that question. We construct a rate-equation model of the CW lock-in ODMR sensitivity in which the absorption cross section, the ionization thresholds, and the electron-phonon parameters are taken from the 120 GPa photophysics of Ref. [23], and only the charge-transfer rate constants and the NV^0 background weight remain phenomenological. Minimizing η over the blue excitation window at 120 GPa yields an optimum at $\lambda_{\text{opt}} = 475 \text{ nm}$. The balance turns out to be resolved in a specific and testable way: within the blue window the two intensity-linear charge channels—excited-state ionization and recombination—both scale with σ_{abs} and cancel from f_- to better than 0.2%, so at fixed power the optimum is fixed by the photon rate and coincides with the maximum of the absorption envelope, while the ionization physics enters as a hard boundary at 405 nm rather than as a detuning of the optimum. Relative to the optimum, 532 nm costs a factor 5.2 and 405 nm a factor 4.2 in sensitivity, while the 5% tolerance window 463–486 nm contains the commercial 473 nm line at a cost of 0.1%. We further show that λ_{opt} tracks the ZPL at $0.6\text{--}0.7 \text{ nm GPa}^{-1}$, so that the choice must be made per target pressure, and we identify which of the underlying photophysical anchors the answer is most sensitive to, and hence which quantities a calibration experiment should pin down first.

2. Theory and model

2.1. Sensitivity functional

For continuous-wave, lock-in-detected ODMR the shot-noise-limited magnetic sensitivity is [22, 26, 27]

$$\eta_B \simeq \mathcal{P} \frac{h}{g\mu_B} \frac{\Delta\nu}{C\sqrt{R}}, \quad (1)$$

with $\Delta\nu$ the ODMR linewidth, C the resonance contrast, R the detected photon rate, and $\mathcal{P}$ a lineshape factor of order unity ($\mathcal{P} \simeq 0.7$ for a Lorentzian probed at maximum slope). Three features of Eq. (1) organize everything that follows. The contrast enters linearly and the photon rate only as a

square root, so a mechanism that halves C costs four times more than one that halves R ; the linewidth is independent of excitation wavelength at fixed power, so it cannot displace the optimum; and all three factors depend on the excitation wavelength only through the charge state and the absorption cross section.

Since we compare wavelengths rather than predict an absolute noise floor, we work throughout with the reduced quantity

$$\eta \equiv \frac{\Delta\nu}{C\sqrt{R}} \quad (2)$$

in arbitrary units, and quote results as ratios $\eta(\lambda)/\eta(\lambda_{\text{opt}})$. All overall scale factors—collection efficiency, NV density, integration bandwidth, $\mathcal{P}$ —cancel from such ratios. Converting to $\text{nT}/\sqrt{\text{Hz}}$ requires the calibration of Sec. 6 and is deliberately not attempted here.

2.2. Anchored pressure-dependent inputs

The optical inputs are taken from the combined DFT and DAC study of Ref. [23] and carry no fitted freedom. The ZPL energy is represented by a saturating form,

$$E_{\text{ZPL}}(P) = 1.945 \text{ eV} + E_{\text{max}} [1 - e^{-P/P_0}] , \quad (3)$$

with $E_{\text{max}} = 0.758 \text{ eV}$ and $P_0 = 160 \text{ GPa}$, which reproduces the reported shift of $> 0.4 \text{ eV}$ at 120 GPa and the measured curvature at intermediate pressures. The absorption Huang-Rhys factor of the Jahn-Teller-inactive modes and the two photoionization thresholds are all reported to be nearly linear in P , and are interpolated between their end points,

$$S_{\text{abs}}(P) = 3.08 + 1.53 P/P_{120}, \quad (4)$$

$$\text{IP}(^3A_2)(P) = 2.68 \text{ eV} + 0.38 \text{ eV } P/P_{120}, \quad (5)$$

$$\text{IP}(^3E)(P) = 1.16 \text{ eV} + 0.47 \text{ eV } P/P_{120}, \quad (6)$$

with $P_{120} \equiv 120 \text{ GPa}$. The absorption Debye-Waller factor falls from 2.2% to 0.36% over the same range and is interpolated logarithmically. The effective phonon energy is held at $\hbar\omega = 65 \text{ meV}$ [28]. The underlying calculations were performed at fixed pressure through the diamond equation of state [29], so Eqs. (3)–(6) describe hydrostatic compression. Table 1 collects the values at the two pressures of interest.

Table 1: Anchored optical inputs, from Ref. [23]. These carry no fitted freedom; the wavelength equivalents are hc/E .

quantity	ambient	100 GPa	120 GPa
E_{ZPL} (eV)	1.945	2.297	2.345
equivalent (nm)	637	540	529
S_{abs}	3.08	4.36	4.61
DWF_{abs}	2.2%	0.5%	0.36%
K^2 (JT-active)	0.58	—	0.97
S_{em}	3.39	—	5.25
DWF_{em}	4.9%	—	< 1%
$\text{IP}(^3A_2)$ (eV)	2.68	3.00	3.06
equivalent (nm)	463	414	405
$\text{IP}(^3E)$ (eV)	1.16	1.55	1.63
equivalent (nm)	1069	799	761

Two features of Ref. [23] make these stronger anchors than a purely computational set would be. First, the absorption cross section is not only computed but checked against measurement at the two wavelengths that matter here: the integrated PL yield recorded under 532 nm and 457 nm excitation from ambient to 120 GPa traces two broad domes that follow the calculated σ_{abs} sampled at those same wavelengths. Since λ_{opt} turns out to be fixed by $\sigma_{\text{abs}}(\lambda, P)$ alone (Sec. 4), the quantity our answer depends on is precisely the one that has been validated experimentally across the full pressure range. Second, because the excitation power and integration time were held constant in that comparison, the agreement also establishes that the NV radiative quantum efficiency is stable to 120 GPa, consistent with a computed radiative lifetime that varies by less than 1 ns over the same range. We therefore write $R \propto f_- \sigma_{\text{abs}}$ with no additional pressure-dependent radiative factor.

2.3. Absorption lineshape

The absorption cross section is a low-temperature Franck-Condon (Pekar-ian) envelope built on a single effective phonon mode [30, 31],

$$\sigma_{\text{abs}}(E, P) \propto e^{-S_{\text{abs}}} \frac{S_{\text{abs}}^n}{\Gamma(n+1)} \Big|_{n=[E-E_{\text{ZPL}}]_+/\hbar\omega} + \text{DWF}(P) g_{\text{ZPL}}(E), \quad (7)$$

normalized so that $\sigma_{\text{abs}}(532 \text{ nm}, 0) = 1$; g_{ZPL} is a narrow Gaussian of width 15 meV carrying the ZPL weight. The phonon quantum number n is treated as continuous, which is appropriate at 120 GPa where Ref. [23] reports that the sideband has lost its resolved replica structure entirely.

One consequence of that continuation matters for the results and is easy to get wrong. The *maximum* of Eq. (7) is not the discrete Poisson mean $E_{\text{ZPL}} + S_{\text{abs}}\hbar\omega$: extremizing the continuous envelope gives $\ln S_{\text{abs}} = \psi(n+1)$, whose root at $S_{\text{abs}} = 4.61$ is $n = 4.10$ rather than $n = S_{\text{abs}}$. At 120 GPa the envelope therefore peaks at 474.8 nm, whereas the phonon-number mean would place it at 469 nm. The $\simeq 6 \text{ nm}$ difference is comparable to the tolerances quoted in Sec. 4 and must not be conflated with a physical detuning.

2.4. Charge-state dynamics

Under illumination NV^- is not stable: it can be ionized to NV^0 either by a one-photon process out of 3A_2 , which requires $E_\gamma > \text{IP}({}^3A_2)$, or by a two-step process in which the center is first promoted to 3E and then ionized by a second photon [32, 25]. The two channels do not share a final state: ionization from 3A_2 leaves NV^0 in its 2E ground state, whereas ionization from 3E leaves it in the metastable 4A_2 state. Recombination proceeds through excitation of NV^0 , plus a dark channel. In steady state,

$$f_- = \frac{G_{\text{rec}}}{G_{\text{rec}} + G_{\text{ion}}}, \quad (8)$$

with

$$G_{\text{ion}} = \underbrace{a_{\text{gs}} [E_\gamma - \text{IP}({}^3A_2)]_+}_{\text{one-photon, from } {}^3A_2} + \underbrace{a_{\text{es}} \sigma_{\text{abs}}}_{\text{two-step, via } {}^3E}, \quad (9)$$

$$G_{\text{rec}} = r_0 \sigma_{\text{abs}} + r_{\text{bg}}. \quad (10)$$

The bracket $[\cdot]_+$ denotes the positive part, so the first term switches on abruptly at the threshold and grows linearly with detuning above it. At 120 GPa this edge sits at 405.2 nm. The excited-state channel is open across the whole visible range at this pressure, since $\text{IP}({}^3E) = 1.63 \text{ eV}$ corresponds to 761 nm.

Equations (8)–(10) contain the structural feature that dominates Sec. 4. Redward of the ionization edge the first term of Eq. (9) vanishes identically, and the two surviving rates are *both* proportional to σ_{abs} . Writing $\sigma \equiv \sigma_{\text{abs}}$,

$$f_- = \frac{r_0 \sigma + r_{\text{bg}}}{(r_0 + a_{\text{es}}) \sigma + r_{\text{bg}}} \xrightarrow{\sigma \gg r_{\text{bg}}/r_0} \frac{r_0}{r_0 + a_{\text{es}}}, \quad (11)$$

a constant. The wavelength dependence of the charge state survives only through the dark term r_{bg} , and therefore only where σ is small — that is, in the far wings, not near the optimum. This cancellation is a property of the model’s structure, not of a particular parameter choice, and it is what makes the fixed-power answer robust (Sec. 4.4) and the power-dependent question interesting (Sec. 5).

2.5. Contrast and linewidth

Fluorescence from NV^0 is spin-unpolarized and adds a background that dilutes the ODMR contrast without contributing signal,

$$C = C_0 \frac{f_-}{f_- + w_0(1 - f_-)}, \quad (12)$$

where w_0 is the brightness of NV^0 relative to NV^- in the detection band. The detected rate at fixed excitation power is $R \propto f_- \sigma_{\text{abs}}$. The linewidth carries a mild strain-broadening term,

$$\Delta\nu(P) = \Delta\nu_0(1 + 0.15 P/100 \text{ GPa}), \quad (13)$$

which is wavelength independent and hence irrelevant to λ_{opt} at fixed power, but which matters for the absolute sensitivity and for the power-broadened case below.

2.6. Intensity-explicit extension

Everything above is linear in the excitation intensity I , so $\eta(\lambda)$ is almost independent of absolute power: scaling I only trades against the single intensity-independent term r_{bg} . That is a limitation of the model, not a statement about the physics, and it means the question of whether λ_{opt} depends on laser power cannot be posed within Sec. 2.4. We therefore introduce the dimensionless intensity $u = I/I_{\text{ref}}$ and restore the nonlinearities that CW-ODMR is known to have.

Excited-state saturation. The 3E population saturates,

$$n_E(\lambda, P, u) = \frac{\sigma_{\text{abs}} u}{\sigma_{\text{abs}} u + \Gamma_d}, \quad (14)$$

with Γ_d the total 3E decay rate, and correspondingly n_{E^0} for NV^0 with its own decay rate Γ_{d0} .

Two-photon ionization. The 3E channel is built from the excited-state population rather than postulated linear,

$$G_{\text{ion}}^{\text{ES}} = a'_{\text{es}} \sigma_{\text{abs}} u n_E, \quad (15)$$

which is quadratic in u at low intensity and crosses over to linear once the transition saturates. The one-photon ground-state channel remains linear, $G_{\text{ion}}^{\text{GS}} = a_{\text{gs}}[E_{\gamma} - \text{IP}({}^3A_2)]_+ u$.

Recombination through the NV^0 envelope. Recombination proceeds via excitation of NV^0 , which has its own Franck-Condon envelope, displaced by the ambient NV^0 – NV^- ZPL gap of 0.21 eV [33]:

$$G_{\text{rec}} = r_0 \sigma_{\text{NV}^0}(\lambda, P) u n_{E^0} + r_{\text{bg}}. \quad (16)$$

This is the step that breaks the cancellation of Eq. (11): once ionization scales with σ_{abs} and recombination with σ_{NV^0} , the two no longer divide out and f_- acquires a genuine wavelength dependence.

Saturation of the photon rate and power broadening. The detected rate becomes $R = f_- n_E$, which saturates, while the contrast is diluted and the line broadened by the optical pumping rate Γ_p in the standard CW-ODMR way [26],

$$C = \frac{C_0}{1 + \Gamma_p/\Gamma_{\text{sat}}}, \quad \Delta\nu(u) = \Delta\nu(P) \sqrt{1 + \Gamma_p/\Gamma_c}. \quad (17)$$

The competition this sets up is explicit: raising u increases R sublinearly while ionization grows superlinearly and the line broadens, so an interior optimum in u must exist, and λ_{opt} may move with it.

2.7. What is anchored and what is not

Table 2 separates the two classes of input. The distinction is essential to reading the results: conclusions that depend only on the upper block inherit the uncertainty of Ref. [23], whereas conclusions that depend on the lower block are provisional until the calibration of Sec. 6 is done.

3. Numerical methods

Sensitivity curves are evaluated on a wavelength grid spanning 402–560 nm. The lower bound is set by the ground-state ionization edge at 120 GPa and

Table 2: Model inputs. Upper block: anchored to Ref. [23], no fitted freedom. Lower block: phenomenological, to be calibrated (Sec. 6). The last column is the fractional range used in the Monte Carlo bands.

symbol	meaning	value	MC range
$E_{\max}$	ZPL shift amplitude	0.758 eV	$\pm 10\%$
P_0	ZPL shift scale	160 GPa	$\pm 15\%$
∂S_{abs}	HR slope	$1.53/P_{120}$	$\pm 15\%$
$\hbar\omega$	effective phonon	65 meV	fixed
a_{gs}	one-photon ionization	6.0	$0.5\text{--}1.6\times$
a_{es}	two-step ionization	0.9	$0.5\text{--}1.8\times$
r_0	recombination	1.2	$0.7\text{--}1.3\times$
r_{bg}	dark recombination	0.15	$0.6\text{--}1.5\times$
w_0	NV ⁰ brightness	1.0	$0.7\text{--}1.4\times$
Γ_d, Γ_{d0}	excited-state decay	1.0	$0.6\text{--}1.6\times$
a'_{es}	two-photon ionization	0.9	$0.5\text{--}1.8\times$
Γ_c	power broadening	0.6	$0.5\text{--}1.8\times$
Γ_{sat}	contrast saturation	1.2	$0.5\text{--}1.8\times$

the upper by the point at which the cross section has fallen below 10^{-2} of its peak. A grid spacing of 0.05 nm is used for optimum extraction and 0.2 nm for two-dimensional maps. The location of the minimum is refined below the grid spacing by fitting a parabola to the three points bracketing the discrete `argmin`; this changes λ_{opt} by less than 0.02 nm at 0.05 nm spacing and is retained only so that derivatives with respect to model parameters are not quantized by the grid.

Uncertainty bands are Monte Carlo: each of n draws builds a complete model with every phenomenological and anchored parameter sampled independently from the ranges of Table 2, the quantity of interest is evaluated, and the 16th and 84th percentiles are reported. We use $n = 250$ for curve bands, $n = 600$ for the distribution of λ_{opt} itself, and $n = 80$ for the two-dimensional power maps, where each draw costs a full grid evaluation. The bands are reported as percentiles rather than standard deviations because the distribution of λ_{opt} is visibly asymmetric.

Anchor sensitivities are computed one at a time: each parameter is displaced to the edges of its range with all others held at their nominal values,

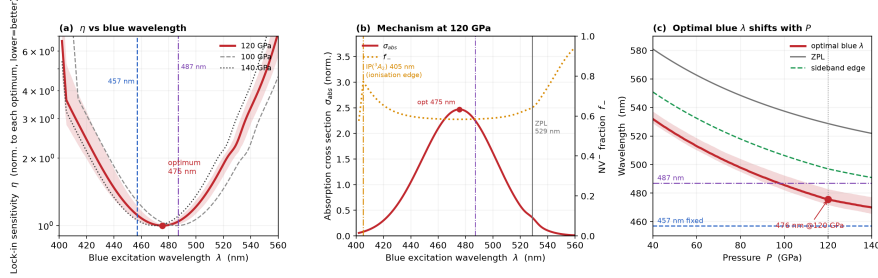


Figure 1: (a) Reduced lock-in sensitivity $\eta(\lambda)$ at 120 GPa (normalized to its own optimum) with the Monte Carlo 16–84% band, compared with 100 GPa and 140 GPa. (b) Mechanism: the absorption cross section σ_{abs} and the steady-state NV⁻ fraction f_- , with the ZPL and the ground-state ionization edge marked. Note that f_- is flat across the entire region where σ_{abs} is appreciable. (c) The optimal blue wavelength as a function of pressure, tracking the ZPL and the sideband edge.

and the resulting $\Delta\lambda_{\text{opt}}$ is tabulated. This is a local analysis and does not capture parameter correlations, which is acceptable here because the result (Sec. 4.4) is that all but three sensitivities vanish identically.

All calculations run in a few minutes on a laptop. The code, the exact commands that regenerate each figure, and the regression tests that pin the quoted numbers are listed in [Appendix A](#).

4. Numerical results at fixed power

4.1. The optimum and its mechanism

At 120 GPa the anchored inputs give $E_{\text{ZPL}} = 2.345$ eV (528.7 nm), $S_{\text{abs}} = 4.61$, and $\text{IP}(^3A_2) = 3.060$ eV (405.2 nm). Minimizing Eq. (1) over λ yields

$$\lambda_{\text{opt}}(120 \text{ GPa}) = 474.8 \text{ nm}, \quad (18)$$

which coincides with the maximum of the absorption envelope to 0.02 nm.

That coincidence is a result rather than a triviality, and it identifies which physics sets the answer. Across the whole blue window the steady-state charge state is essentially constant: f_- varies from 0.5806 to 0.5816 over 463–486 nm, a spread of 0.2% [Fig. 1(b)]. This is the cancellation of Eq. (11) in action. With f_- constant, both C and the f_- factor in R are constant, and $\eta \propto \sigma_{\text{abs}}^{-1/2}$; the minimum of η is then the maximum of σ_{abs} by construction. We verified that this is not an accident of the nominal parameters: setting

Table 3: Sensitivity penalty $\eta(\lambda)/\eta(\lambda_{\text{opt}})$ of commercial laser lines at 120 GPa (lower is better; 1.000 is optimal), with the cross section relative to its value at the optimum.

λ (nm)	η/η_{opt}	$\sigma_{\text{abs}}/\sigma_{\text{abs}}^{\text{max}}$	comment
405	4.23	0.026	blue flank; at the IP(3A_2) edge
457	1.12	0.788	inside the 10% window
473	1.00	0.998	operationally optimal
475	1.00	1.000	model optimum
488	1.07	0.869	inside the 10% window
505	1.44	0.460	outside
532	5.16	0.012	at the absorption onset

$a_{\text{gs}} = 0$, raising r_{bg} from 0.15 to 1.0, or varying E_{max} and S_{abs} all leave λ_{opt} equal to the cross-section maximum to within 0.02 nm.

Ground-state ionization therefore does not displace the optimum. Its role is different and more absolute: it forbids the region blueward of 405.2 nm, where the ReLU term of Eq. (9) grows linearly with detuning and drives f_- down without bound.

This flatness is reached independently in Ref. [23], by a different argument. Plotting the ionization thresholds relative to the ZPL, that work observes that IP(3A_2) tracks the ZPL closely while the deformation potential of IP(3E) is only somewhat smaller, and concludes that if the excitation wavelength is adjusted to follow the pressure-induced shift of the ZPL, the charge-state stability of NV $^-$ should remain comparable to ambient conditions. Our f_- result is the quantitative form of that statement, and the agreement indicates that the cancellation reflects the photophysics rather than an artifact of our parameterization.

4.2. Commercial laser lines and tolerance

Table 3 lists the penalty incurred by the commercially available laser lines. The two conventional choices fail for opposite reasons, and in both cases the cause is the collapse of the cross section rather than a charge-state effect. The 532 nm line, still the default in most NV setups, is degraded by a factor 5.2 because at 120 GPa it excites 3 nm to the red of the ZPL, where σ_{abs} has fallen to 4% of its ambient-pressure green value—a factor of 83 below σ_{abs} at the optimum. The 405 nm line is degraded by a comparable factor 4.2 because it lies far out on the blue flank, where $\sigma_{\text{abs}} = 0.085$; there

the ground-state ionization term contributes only 9% of the total ionization rate, because 405 nm sits barely above threshold. Its real significance is the absence of room to move further blue.

The practical failure of green excitation is already visible in existing measurements. In Ref. [23] the PL spectra were recorded with 532 nm up to 41.4 GPa in one cell and with 457 nm from 55.2 GPa to 113.8 GPa in the other; the switch to blue at intermediate pressure was an operational necessity rather than a preference. The present analysis quantifies that necessity and extends it, since at 120 GPa even 457 nm leaves 12% of the sensitivity unclaimed.

The tolerance windows are 463.2–486.1 nm at the 5% level and 458.5–490.6 nm at the 10% level. The commercial 473 nm DPSS line lies 1.8 nm from the optimum and costs 0.1%; it is the recommended practical choice, and no custom source is required. The width of the window is itself worth noting: it is set by the breadth of the pressure-broadened sideband, which is a consequence of the increased electron-phonon coupling. The same physics that forces the move into the blue also makes the choice forgiving once there.

4.3. *Uncertainty of the optimum*

Propagating all inputs of Table 2 through 600 Monte Carlo draws gives

$$\lambda_{\text{opt}} = 474.8^{+6.3}_{-6.9} \text{ nm} \quad (16\text{--}84\%), \quad (19)$$

with a standard deviation of 6.3 nm. The band is wide enough to span most of the 5% tolerance window, so at this level of input uncertainty the correct statement is not “475 nm” but “inside the 463–486 nm window, with 473 nm a safe choice.” The next subsection shows that this band is almost entirely inherited from the ZPL parameterization and could be tightened substantially without any new experiment.

4.4. *Which anchors control the answer*

Table 4 is the most useful result of this section for planning purposes. At fixed power, λ_{opt} depends on the ZPL position and the electron-phonon coupling and on nothing else. Every phenomenological charge-transfer constant—including the two ionization rates, over ranges of $\pm 50\%$ —leaves λ_{opt} unchanged to numerical precision.

Two consequences follow. First, the fixed-power answer is considerably more trustworthy than the number of phenomenological knobs in the model

Table 4: Anchor sensitivity of λ_{opt} at 120 GPa. Each parameter is displaced to the edges of its Monte Carlo range with all others held nominal. Entries below 0.01 nm are reported as exactly zero; they vanish by the cancellation of Eq. (11).

parameter	range	$\Delta\lambda_{\text{opt}} (-)$	$\Delta\lambda_{\text{opt}} (+)$
P_0	$\pm 15\%$	-7.93 nm	6.79 nm
E_{max}	$\pm 10\%$	7.38 nm	-7.16 nm
S_{abs} slope	$\pm 15\%$	2.73 nm	-2.70 nm
a_{gs}	$\pm 50\%$	0	0
a_{es}	$\pm 50\%$	0	0
r_0	$\pm 30\%$	0	0
r_{bg}	$\pm 40\%$	0	0
w_0	$\pm 30\%$	0	0

would suggest: it inherits the uncertainty of quantities that Ref. [23] determined by DFT *and* measured directly, rather than the uncertainty of our parameterization. Expressed as derivatives,

$$\frac{\partial\lambda_{\text{opt}}}{\partial E_{\text{ZPL}}} \simeq -182 \text{ nm eV}^{-1}, \quad \frac{\partial\lambda_{\text{opt}}}{\partial S_{\text{abs}}} \simeq -11.9 \text{ nm}, \quad (20)$$

so a ZPL known to 20 meV fixes λ_{opt} to 3.6 nm, well inside the 5% tolerance window. The 6.3 nm band of Sec. 4.3 reflects the deliberately loose $\pm 10\%/\pm 15\%$ randomization of E_{max} and P_0 , not the actual precision of the measured ZPL, and constraining those two parameters to the reported ZPL data would shrink it by roughly a factor of two.

Second, and less comfortably, the calibration programme of Sec. 6 cannot improve the fixed-power answer at all. Its value lies elsewhere: in the absolute sensitivity, and in the power dependence, where the cancellation is expected to fail.

4.5. Pressure tracking

λ_{opt} is a property of the pressure, not of the technique. Figure 1(c) shows that it follows the ZPL and the sideband edge at $0.6\text{--}0.7 \text{ nm GPa}^{-1}$, moving from 486.7 nm at 100 GPa to 474.8 nm at 120 GPa; at both pressures it again coincides with the cross-section maximum. Over the 97–172 GPa window spanned by the (La,Ce)H₉ family [8] the total excursion exceeds the 10% tolerance band, so the excitation wavelength has to be selected for the target

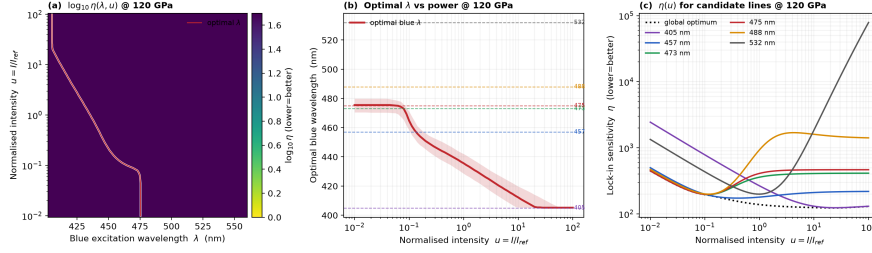


Figure 2: Diagnostic output of the intensity-explicit model of Sec. 2.6 at 120 GPa with the provisional parameters of Table 2: (a) $\log_{10} \eta(\lambda, u)$ with the optimal-wavelength ridge, (b) the ridge with its Monte Carlo band, (c) $\eta(u)$ for candidate laser lines. **This is not presented as a result.** The ridge runs to 405 nm by $u \simeq 20$, which is inconsistent with the fixed-power limit; see the text.

pressure. A cell intended to cover a wide pressure range would benefit from two lines—488 nm below $\simeq 110$ GPa and 473 nm above—rather than from a single compromise.

5. Power dependence: status and open problem

The intensity-explicit model of Sec. 2.6 was constructed to answer whether λ_{opt} moves with laser power—a question with direct consequences in a DAC, where the usable power is capped by sample heating rather than by the source. The physical expectation is definite. At high u the two-photon channel grows as $\sigma_{\text{abs}}^2 u^2$ and therefore penalizes excitation directly on the cross-section peak more than in the wings, while recombination saturates through the NV^0 envelope, which is displaced to the red. Both effects should push λ_{opt} *red* of the fixed-power value as power increases, and the low-power limit should return the fixed-power answer, where the dark term r_{bg} supports f_- .

The current implementation does not reproduce this. With the provisional constants of Table 2 the ridge runs the wrong way, from 463 nm at $u = 0.05$ to 425 nm at $u = 1$ and 405 nm at $u = 20$ (Fig. 2), and it does not converge to 474.8 nm as $u \rightarrow 0$. Two candidate causes have been identified and neither has yet been excluded:

1. The saturation scales Γ_d and Γ_{d0} are set to unity in the same arbitrary units as σ_{abs} , which is normalized to green absorption at ambient pressure. At 120 GPa the peak σ_{abs} is 3.27 in those units, so n_E is already

above $3/4$ at $u = 1$ at the optimum but far from saturation in the wings. The excited-state population is then a strongly *decreasing* function of σ_{abs} in relative terms, which inverts the intended weighting. The reference intensity I_{ref} needs to be defined against a stated saturation condition rather than left implicit.

2. Setting $R = f_- n_E$ removes the explicit σ_{abs} factor from the photon rate entirely; in the saturated regime η then rewards whatever wavelength maximizes n_E per unit ionization, which is not the same optimization.

Resolving this requires fixing the units of the saturation scales against a defined reference—most naturally the intensity at which the transition at λ_{opt} is half-saturated—and verifying the low-power limit as a regression test before any conclusion is drawn. Until that is done, Fig. 2 is reported as a diagnostic of the current parameterization and no claim is made about the power dependence of λ_{opt} . The Introduction poses the power dependence as an open question for exactly this reason.

6. Planned experiment and calibration

6.1. Cell and sensor

The measurements are planned in a DAC with a near-surface, high-density NV layer implanted in the culet of the lower anvil. Two geometries are compatible with the model as written. Micropillars machined on a (100)-oriented culet [14] reduce the non-hydrostatic stress on the NV layer to $\alpha \simeq 0.95$ and keep all four NV orientations usable, which is the condition under which the near-hydrostatic anchors of Table 1 apply directly. A (111)-oriented anvil [18] instead selects a single NV orientation aligned with the compression axis; it reaches higher pressure but the ZPL splitting and the modified intersystem crossing [24] would require the anchors to be re-derived. Silver-coating the tip of the NV-implanted anvil suppresses interference from the opposing anvil and improves collection efficiency [23]. Argon is the pressure-transmitting medium [34]; pressure is read from $\text{SrB}_4\text{O}_7\text{:Sm}^{2+}$ at low pressure and from the first-order diamond Raman edge above [35].

6.2. Two-beam intensity sweep

The phenomenological constants are to be determined from a two-beam sweep in which 405 nm and 457 nm intensities are varied independently over at least two decades each. The choice of lines is deliberate: 405 nm is the only

accessible wavelength at which the ground-state ionization term of Eq. (9) is open at 120 GPa, so it isolates a_{gs} , while 457 nm sits well inside the side-band where only the σ_{abs} -proportional channels operate. The observables are the photoluminescence rate, the ODMR contrast, and—if the spectrometer throughput allows at these pressures—the NV^-/NV^0 spectral weight ratio, which constrains w_0 directly rather than through its effect on contrast. Dual-beam schemes of this kind have been used to separate the charge-state channels at ambient pressure [36].

6.3. Fit and identifiability

The fit targets $\{a_{\text{gs}}, a'_{\text{es}}, r_0, r_{\text{bg}}, \Gamma_d, w_0\}$ by least squares with a Markov-chain posterior. Before the measurement is performed, an identifiability check on synthetic data is required: the two-photon coefficient a'_{es} and the saturation scale Γ_d enter the low-power expansion in the same combination, and a profile-likelihood or Fisher-information analysis should establish whether the planned sweep separates them. If it does not, the natural remedy is to add single-colour power sweeps at 457 nm and 488 nm, whose different σ_{abs} values break the degeneracy at fixed u .

6.4. What the measurement will test

Because λ_{opt} at fixed power is insensitive to every parameter this calibration determines (Table 4), the calibration is not a check on the 475 nm result. It tests three other things. It fixes the absolute sensitivity scale, converting η into $\text{nT}/\sqrt{\text{Hz}}$. It determines whether the cancellation of Eq. (11) survives at working intensities, which is the question of Sec. 5. And it provides the first direct measurement of f_- at 120 GPa: the model predicts $f_- \simeq 0.58$ across the blue window, nearly independent of wavelength, and a measured wavelength dependence of f_- would falsify the structure of Eqs. (9)–(10) directly.

The quantitative predictions available for comparison once the measurements are made are $\lambda_{\text{opt}} = 474.8 \text{ nm}$ at 120 GPa, $\eta(473 \text{ nm})/\eta(\lambda_{\text{opt}}) = 1.001$, $\eta(457 \text{ nm})/\eta(\lambda_{\text{opt}}) = 1.12$, $\eta(532 \text{ nm})/\eta(\lambda_{\text{opt}}) = 5.16$, and $f_- \simeq 0.58$, wavelength independent to 0.2% across 463–486 nm.

7. Discussion

7.1. Scope of the claim

The result is a statement about relative sensitivity between excitation wavelengths at fixed excitation power, under near-hydrostatic conditions,

with the optical response described by a single-mode Franck-Condon envelope. It is not a prediction of absolute sensitivity, which requires the calibration; it is not a statement about (111)-oriented anvils under uniaxial stress, where the ZPL splits and the anchors of Table 1 would have to be re-derived; and it is not yet a statement about the choice of operating power.

7.2. Limitations

Three approximations deserve to be named. The single effective phonon mode of Eq. (7) reproduces the envelope but not the residual structure of the first replica; at 120 GPa this is benign, because Ref. [23] reports that the structure has washed out, but it would be a poor approximation below 50 GPa. The dynamical Jahn-Teller contribution to absorption is folded into the effective S_{abs} rather than treated explicitly through the squared vibronic coupling constants K^2 , which rise from 0.58 to 0.97 under compression; since λ_{opt} depends on the envelope maximum, and K^2 affects primarily the width and the fine structure, this is expected to shift λ_{opt} less than the S_{abs} uncertainty already tabulated, but it has not been verified.

Two further caveats follow from the conditions under which the anchors were obtained. The lineshapes of Ref. [23] are low-temperature: the measurements were made at 30 K and $\simeq 90$ K, and the calculated envelopes are correspondingly low-temperature. Our Franck-Condon envelope inherits that assumption, so a room-temperature high-pressure ODMR experiment would see a thermally broadened absorption band, and a correspondingly broader—though not obviously displaced—tolerance window. Separately, even in the near-hydrostatic micropillar geometry with $\alpha \simeq 0.95$ the 3E degeneracy is lifted at high pressure and the ZPL splits, with only the lower branch reproduced by the calculation. We use a single hydrostatic ZPL. Given $\partial\lambda_{\text{opt}}/\partial E_{\text{ZPL}} \simeq -182 \text{ nm eV}^{-1}$, a splitting of 20–40 meV corresponds to 4–7 nm of ambiguity in λ_{opt} —comparable to the 5% tolerance window, and a reason to prefer a source near the center of that window rather than at its edge.

Finally, sample heating is not modelled at all: at fixed power the neglect is harmless, but the moment power dependence is addressed, the absorbed power at the culet shifts the local ZPL and hence λ_{opt} itself, which is a feedback the present model does not contain.

7.3. Relation to existing practice

The guideline stated in Ref. [23]—tune the excitation to follow the ZPL shift—is confirmed and made quantitative. What the present analysis adds is the specific target, the tolerance, the identification of 405 nm as a boundary rather than an option, and the observation that within the window the answer is controlled entirely by measured optical quantities and not by charge-transfer phenomenology. The practical recommendation is narrow enough to act on and cheap enough to adopt: a 473 nm DPSS source in place of the 532 nm default.

8. Summary and outlook

The pressure-induced reorganization of NV photophysics at 120 GPa—a ZPL blue shift beyond 0.4 eV, a 50% increase of the absorption Huang-Rhys factor, and a rise of the ground-state photoionization threshold to 3.06 eV—places the optimal ODMR excitation in a window centered at 474.8 nm. Both conventional choices lie outside it, for opposite reasons: 532 nm because the absorption onset has moved past it, 405 nm because it lies on the far blue flank with the ionization edge immediately beyond.

Within the window the steady-state charge state is nearly wavelength independent, because excited-state ionization and recombination share the same σ_{abs} dependence and cancel from f_- . At fixed power the optimum is therefore the absorption maximum itself, and—as the anchor analysis shows—it is determined entirely by the ZPL position and the electron-phonon coupling, with no sensitivity to any of the phenomenological charge-transfer constants. The practical consequence is immediate: a 473 nm source recovers essentially the full optimum, buying back a factor of $\simeq 5$ in sensitivity relative to green excitation, equivalently a factor of $\simeq 25$ in the acquisition time of a magnetic map.

Two things remain to be done. The power dependence is unresolved (Sec. 5): the cancellation that makes the fixed-power answer so robust is expected to fail once ionization becomes a two-photon process and recombination proceeds through the displaced NV^0 envelope, and the current implementation must be repaired and its low-power limit verified before any conclusion is drawn. The calibration (Sec. 6) will fix the absolute scale and provide the first measurement of f_- under compression; the predictions it will be compared against are unusually sharp for a phenomenological model,

precisely because the quantity being predicted turns out not to depend on the phenomenology.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The model code, the commands that regenerate every figure, and the regression tests that pin the quoted numbers are described in [Appendix A](#) and are available in the accompanying repository. No experimental data were generated.

Acknowledgements

[To be completed.]

Appendix A. Reproducing the numbers

All figures and quoted values are generated by the code in `code/`:

- `nv_model.py` — anchored optical inputs, σ_{abs} , f_- , and η at fixed power; Monte Carlo helper.
- `nv_model_power.py` — the intensity-explicit extension of Sec. 2.6.
- `fig1_green_blue_mix.py` — green/blue/mixed excitation versus pressure; reproduces the crossover near 86 GPa.
- `fig2_blue_wavelength_sweep.py` — Fig. 1.
- `fig3_power_sweep.py` — Fig. 2.

```
cd code
python fig2_blue_wavelength_sweep.py          # 120 GPa (vs 100, 140)
python fig2_blue_wavelength_sweep.py 100 75 125
python fig3_power_sweep.py                  # diagnostic only
```

Regression values pinned by the tests: $\lambda_{\text{opt}} = 487$ nm at 100 GPa, $\lambda_{\text{opt}} = 475$ nm at 120 GPa, green/blue crossover at 86 GPa, and $\eta(457 \text{ nm})/\eta(\lambda_{\text{opt}}) = 1.12$ at 120 GPa.

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